DAA EXPERIMENT 6

Champagne Tower

Aim

To determine the amount of champagne in a specified glass of a champagne tower.

Algorithm

1. Start with the poured amount in the top glass.
2. Traverse the tower row by row.
3. If a glass contains more than 1 unit, calculate the excess amount.
4. Divide the excess equally between the two glasses in the next row.
5. Continue until the required row is processed.
6. Display the amount in the specified glass, limited to a maximum of 1 unit.
7. Stop.

Program

```
# Champagne Tower

poured = float(input("Enter amount poured: "))
row = int(input("Enter row number (starting from 0): "))
glass = int(input("Enter glass number (starting from 0): "))

tower = [[0.0] * 10 for _ in range(10)]
tower[0][0] = poured

for i in range(row):
    for j in range(i + 1):
        if tower[i][j] > 1:
            extra = (tower[i][j] - 1) / 2
            tower[i + 1][j] += extra
            tower[i + 1][j + 1] += extra

print("Amount of champagne:", min(1.0, tower[row][glass]))
```

Complexity Analysis

- Time Complexity: $O(r^2)$, where r is the requested row
- Space Complexity: $O(r^2)$ for the tower storage

Output

```
Enter row number: 2
Enter glass number: 2
Champagne in the glass: 0
PS C:\Users\jithe\OneDrive\Documents\DAA - programs> 1
```

Result

Thus, the amount of champagne in the specified glass was successfully calculated.